

Exercise JA

Q1. If $y(x)$ satisfies the differential equation $y' - y \tan x = 2x \sec x$ and $y(0) = 0$, then :

(A) $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{8\sqrt{2}}$ (B) $y'\left(\frac{\pi}{4}\right) = \frac{\pi^2}{18}$ (C) $y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{9}$ (D) $y'\left(\frac{\pi}{3}\right) = \frac{4\pi}{3} + \frac{2\pi^2}{3\sqrt{3}}$

Ans. D, A

Sol. -

Q2. A curve passes through the point $\left(1, \frac{\pi}{6}\right)$. Let the slope of the curve at each point (x, y) be $\frac{y}{x} + \sec\left(\frac{y}{x}\right)$, $x > 0$. Then the equation of the curve is

(A) $\sin\left(\frac{y}{x}\right) = \log x + \frac{1}{2}$ (B) $\cos ec\left(\frac{y}{x}\right) = \log x + 2$ (C) $\sec\left(\frac{2y}{x}\right) = \log x + 2$
 (D) $\cos\left(\frac{2y}{x}\right) = \log x - 2$

Ans. A

Sol. -

Q3. The function $y = f(x)$ is the solution of the differential equation

$$\frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^4 + 2x}{\sqrt{1 - x^2}} \text{ in } (-1, 1) \text{ satisfying } f(0) = 0. \text{ Then : } \int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x) dx \text{ is :}$$

(A) $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$ (B) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$ (C) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

Ans. B

Sol. -

Q4. Let $y(x)$ be a solution of the differential equation $(1+e^x)y' + ye^x = 1$. If $y(0) = 2$, then which of the following statements is (are) true ?

(A) $y(-4) = 0$ (B) $y(-2) = 0$ (C) $y(x)$ has a critical point in the interval $(-1, 0)$
 (D) $y(x)$ has no critical point in the interval $(-1, 0)$

Ans. A, C

Sol. $(1 + e^x) \frac{dy}{dx} + ye^x = 1$

$$\frac{dy}{dx} + \frac{e^x}{1 + e^x} y = \frac{1}{1 + e^x}$$

$$\text{I.F.} = e^{\int \frac{e^x}{1+e^x} dx} = e^{\ln(1+e^x)} = 1 + e^x$$

complete solution

$$y \cdot (1 + e^x) = \int 1 dx$$

$$(1 + e^x)y = x + c$$

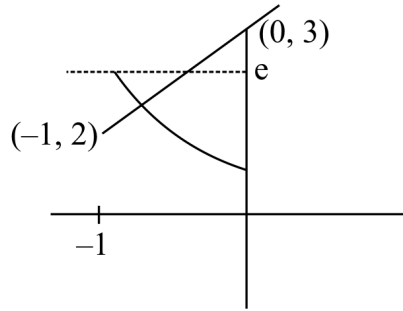
$$x = 0, y = 2 \Rightarrow c = 4$$

$$(1 + e^x)y = x + 4$$

$$y = \frac{x + 4}{e^x + 1}$$

$$x = -4, y = 0$$

$$x = -2, y = \frac{2}{e^{-2} + 1}$$



$$\frac{dy}{dx} = \frac{(e^x + 1) \cdot 1 - (x + 4)e^x}{(e^x + 1)^2}$$

$$\frac{e^x(-x - 3) + 1}{(e^x + 1)^2}$$

$$\frac{dy}{dx} = 0 \Rightarrow x + 3 = e^{-x}$$

$$e^x = \frac{1}{x + 3}$$

- Q5. Consider the family of all circles whose centers lie on the straight line $y = x$. If this family of circles is represented by the differential equation $Py'' + Qy' + 1 = 0$, where P, Q are functions of x, y and y' (here $y' = \frac{dy}{dx}, y'' = \frac{d^2y}{dx^2}, y'' = \frac{d^2y}{dx^2}$), then which of the following statements is (are) true ?

- (A) $P = y + x$ (B) $P = y - x$ (C) $P + Q = 1 - x + y + y' + (y')^2$
 (D) $P - Q = x + y - y' - (y')^2$

Ans. B, C

Sol. -

- Q6. A solution curve of the differential equation $(x^2 + xy + 4x + 2y + 4) \frac{dy}{dx} - y^2 = 0, x > 0$, passes through the point $(1, 3)$. Then the solution curve

- (A) intersects $y = x + 2$ exactly at one point
 (B) intersects $y = x + 2$ exactly at two points (C) intersects $y = (x + 2)^2$

(D) does NOT intersect $y = (x + 3)^2$

Ans. D, A

Sol. $\left[(x+2)^2 + y(x+2)\right] \frac{dy}{dx} - y^2 = 0$

Put $x + 2 = t$

$$\frac{dt}{dy} = \frac{t^2 + ty}{y^2} \text{ which is homogeneous differential equations}$$

By solving it

$$\ln y = \frac{-y}{x+2} + C, C = 1 + \ln 3 \text{ since } y(1) = 3$$

$$\Rightarrow \ln\left(\frac{y}{3}\right) + \frac{y}{x+2} = 1$$

Now : Verify the alternatives.

Q7. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that $f'(x) = 2 - \frac{f(x)}{x}$ for all $x \in (0, \infty)$ and $f(1) \neq 1$. Then

$$(A) \lim_{x \rightarrow 0^+} f'\left(\frac{1}{x}\right) = 1 \quad (B) \lim_{x \rightarrow 0^+} x f'\left(\frac{1}{x}\right) = 2 \quad (C) \lim_{x \rightarrow 0^+} x^2 f'(x) = 0$$

$$(D) |f(x)| \leq 2 \text{ for all } x \in (0, 2)$$

Ans. A

Sol. $f'(x) = 2 - \frac{f(x)}{x} \text{ or } \frac{dy}{dx} + \frac{y}{x} = 2$

$$I.F. = e^{\int \frac{1}{x} dx} = x$$

$$\Rightarrow y \cdot x = x^2 + c \Rightarrow f(x) = x + \frac{c}{x}, c \neq 0 \text{ as } f(1) \neq 1$$

$$(A) \lim_{x \rightarrow 0^+} f'\left(\frac{1}{x}\right) = \lim_{x \rightarrow 0^+} (1 - cx^2) = 1$$

$$(B) \lim_{x \rightarrow 0^+} \left(x f\left(\frac{1}{x}\right)\right) = \lim_{x \rightarrow 0^+} (1 + cx^2) = 1$$

$$(C) \lim_{x \rightarrow 0^+} (x^2 f'(x)) = \lim_{x \rightarrow 0^+} \left(x^2 \left(1 - \frac{c}{x^2}\right)\right) = -C \neq 0$$

$$(D) \lim_{x \rightarrow 0^+} f(x) = +\infty \text{ or } -\infty, \text{ so for } c \neq 0 \text{ } f(x) \text{ is unbounded function for } x \in (0, 2)$$

Q8. If $y = y(x)$ satisfies the differential equation $8\sqrt{x} \left(\sqrt{9 + \sqrt{x}} \right) dy = \left(\sqrt{4 + \sqrt{9 + \sqrt{x}}} \right)^{-1} dx$, $x > 0$ and $y(0) = \sqrt{7}$, then $y(256) =$

$$(A) 80 \quad (B) 9 \quad (C) 16 \quad (D) 3$$

Ans. D

Sol.

$$\frac{dy}{dx} = \frac{\left(\sqrt{4 + \sqrt{9 + x}}\right)^{-1}}{8\sqrt{x}\sqrt{9 + \sqrt{x}}}$$

$$dy = \frac{1}{\sqrt{4 + \sqrt{9 + \sqrt{x}}}} \cdot \frac{1}{\sqrt{9 + \sqrt{x}}} \cdot \frac{1}{8\sqrt{x}} dx$$

$$\text{Let } 4 + \sqrt{9 + \sqrt{x}} = t \Rightarrow \frac{1}{2\sqrt{9 + \sqrt{x}}} \times \frac{1}{2\sqrt{2}} dx = dt$$

$$\int dy = \int \frac{1}{\sqrt{t}} \cdot \frac{1}{2} dt$$

$$y = \sqrt{t} + c$$

$$y = \sqrt{4 + \sqrt{9 + \sqrt{x}}} + c$$

$$\text{at } x = 0, y = \sqrt{7}$$

$$\sqrt{7} = \sqrt{7} + c \Rightarrow c = 0$$

$$y = \sqrt{4 + \sqrt{9 + \sqrt{x}}}$$

$$\text{at } x = 256 \Rightarrow y = \sqrt{4 + \sqrt{9 + \sqrt{256}}} = 3$$

Q9. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function such that $f'(x) > 2f(x)$ for all $x \in \mathbb{R}$, and $f(0) = 1$, then :

(A) $f(x)$ is increasing in $(0, \infty)$ (B) $f'(x) < e^{2x}$ in $(0, \infty)$ (C) $f(x) > e^{2x}$ in $(0, \infty)$

(D) $f(x)$ is decreasing in $(0, \infty)$

Ans. A, C

Sol. $f'(x) - 2f(x) > 0$

$$\frac{d}{dx}(f(x) \cdot e^{-2x}) > 0 \quad g(x) = f(x) \cdot e^{-2x} \text{ is an increasing function.}$$

$$\text{for } x > 0, \quad g(x) > g(0)$$

$$f(x) \cdot e^{-2x} > 1 \quad f(x) > e^{2x}$$

$$\text{Now } f'(x) > 2f(x) > 2 \cdot e^{2x}$$

$\therefore f(x)$ is an increasing function

Q10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be two non-constant differentiable functions. If $f'(x) = (e^{(f(x)-g(x))}) g'(x)$ for all $x \in \mathbb{R}$ and $f(1) = g(2) = 1$, then which of the following statement(s) is (are) TRUE ?

- (A) $f(2) < 1 - \log_e 2$ (B) $f(2) > 1 - \log_e 2$ (C) $g(1) > 1 - \log_e 2$
 (D) $g(1) < 1 - \log_e 2$

Ans. B, C

Sol. $f'(x) = e^{f(x)-g(x)} g'(x) : f(1) = g(2) = 1$

$$e^{-f(x)} = e^{-g(x)} + c$$

$$e^{-f(x)} \cdot f'(x) = e^{-g(x)} \cdot g'(x)$$

$$\int d(e^{-f(x)}) = \int d(e^{-g(x)})$$

$$e^{-f(x)} = e^{-g(x)} + c$$

$$x = 1 \quad \frac{1}{e} = e^{-g(1)} + c$$

$$x = 2 \quad e^{-f(2)} = \frac{1}{e} + c$$

$$e^{-1} - e^{-f(2)} = e^{-g(1)} - e^{-1}$$

$$\Rightarrow e^{-g(1)} + e^{-f(2)} = 2e^{-1}$$

$$e^{-g(1)} < 2e^{-1}$$

$$-g(1) < \ln 2 - 1$$

$$\therefore g(1) > 1 - \ln 2$$

$$\text{Similarly } e^{-f(2)} = 2e^{-1} - e^{-g(1)}$$

$$f(2) > 1 - \ln 2$$

Q11. Let $f : (0, \pi) \rightarrow R$ be a twice differentiable function such that

$$\lim_{t \rightarrow x} \frac{f(x) \sin t - f(t) \sin x}{t - x} = \sin^2 x \text{ for all } x \in (0, \pi).$$

If $f\left(\frac{\pi}{6}\right) = -\frac{\pi}{12}$, then which of the following statement(s) is (are) TRUE ?

(A) $f\left(\frac{\pi}{4}\right) = \frac{\pi}{4\sqrt{2}}$ (B) $f(x) < \frac{x^4}{6} - x^2$ for all $x \in (0, \pi)$

(C) There exists $\alpha \in (0, \pi)$ such that $f'(\alpha) = 0$ (D) $f''\left(\frac{\pi}{2}\right) + f\left(\frac{\pi}{2}\right) = 0$

Ans. B, C, D

Sol. $\lim_{t \rightarrow x} \frac{f(x) \sin t - f(t) \sin x}{t - x} = \sin^2 x$

$$\frac{f(x) \cos x - f'(x) \sin x}{\sin^2 x} = 1$$

$$-d\left(\frac{f(x)}{\sin x}\right) = 1$$

$$\frac{f(x)}{\sin x} = -x + c$$

$$\therefore f\left(\frac{\pi}{6}\right) = -\frac{\pi}{12} \Rightarrow c = 0 \Rightarrow f(x) = -x \sin x$$

(D) $f(x) + f''(x) = -2\cos x$

$$f\left(\frac{\pi}{2}\right) + f''\left(\frac{\pi}{2}\right) = 0$$

$$(A) f\left(\frac{\pi}{4}\right) = \left(-\frac{\pi}{4\sqrt{2}}\right)$$

(C) $f(x)$ is continuous and differentiable and $f(0) = f(\pi) = 0$

Using LMVT $f'(c) = 0$ for some $c \in (0, \pi)$

$$(B) g(x) = -x \sin x + x^2 - \frac{x^4}{6}$$

$$g'(x) = f'(x) + 2x - \frac{2x^3}{3}$$

$$g''(x) = f''(x) + 2x - 2x^2$$

$$g'''(x) = 3 \sin x + x \cos x - 4x = 3(\sin x - x) + x(\cos x - 1)$$

$$\Rightarrow g'''(x) < 0 \Rightarrow g''(x) \text{ is decreasing}$$

$$\text{for } x > 0 \quad g''(x) < g''(0) \Rightarrow g''(x) < 0$$

hence $g'(x)$ is decreasing

$$\text{for } x > 0 \quad g'(x) < g'(0) \Rightarrow g'(x) < 0$$

hence $g(x) < 0$

$$\text{for } x > 0 \quad g(x) < g(0) \Rightarrow g(x) < 0$$

$$\text{Hence } f(x) < \frac{x^4}{6} - x^2 \forall x \in (0, \pi)$$

Q12. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 0$. If $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} = (2 + 5y)(5y - 2)$, then the value of $\lim_{x \rightarrow \infty} f(x)$ is _____.

Ans. 0.4

$$\text{Sol. } \frac{dy}{dx} = (5y + 2)(5y - 2)$$

$$\frac{1}{25} \int \frac{dy}{\left(y + \frac{2}{5}\right)\left(y - \frac{2}{5}\right)} = \int dx$$

$$\frac{1}{25} \cdot \frac{5}{4} \ln \left| \frac{y - \frac{2}{5}}{y + \frac{2}{5}} \right| = x + c$$

$$\frac{1}{20} \ln \left| \frac{5y - 2}{5y + 2} \right| = x + c$$

$$\text{at } x = 0, y = 0 \Rightarrow c = 0$$

$$\text{Hence } \frac{2 - 5y}{2 + 5y} = e^{20x}$$

$$\frac{2 - 5y}{2 + 5y} = e^{20x}, \quad \lim_{x \rightarrow -\infty} e^{20x} = 0$$

$$\Rightarrow \lim_{x \rightarrow -\infty} y = \frac{2}{5} = 0.4$$

Q13. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 1$ and satisfying the equation $f(x + y) = f(x)f'(y) + f'(x)f(y)$ for all $x, y \in \mathbb{R}$

Then, the value of $\log_e(f(4))$ is _____.

Ans. 2

Sol. $f(x+y) = f(x).f'(y) + f'(x).f(y)$

substituting $x = y = 0$, we get

$$f(0) = 2f'(0)$$

$$\Rightarrow f'(0) = \frac{1}{2} \text{ Now substituting } y = 0$$

$$f(x) = f(x).f'(0) + f'(x).f(0)$$

$$\Rightarrow f'(x) = \frac{f(x)}{2}$$

$$\Rightarrow f(x) = ke^{x/2}$$

$$\Rightarrow f(x) = e^{x/2} (\text{as } f(0) = 1)$$

$$\text{Now } \ln(f(x)) = \frac{x}{2} \Rightarrow \ln(f(4)) = 2$$

Q14. Let Γ denote a curve $y = y(x)$ which is in the first quadrant and let the point $(1,0)$ on it. Let the tangent to Γ at a point P intersect the y -axis at Y_p . If PY_p has length 1 for each point P on Γ , then which of the following options is/are correct

$$(A) xy' + \sqrt{1-x^2} = 0 \quad (B) y = -\log_e \left(\frac{1 + \sqrt{1-x^2}}{x} \right) + \sqrt{1-x^2}$$

$$(C) y = \log_e \left(\frac{1 + \sqrt{1-x^2}}{x} \right) - \sqrt{1-x^2} \quad (D) xy' - \sqrt{1-x^2} = 0$$

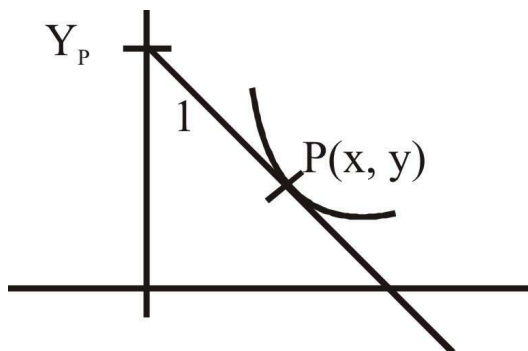
Ans. C, A

Sol. Equation of tangent at P

$$Y - y = m(X - x)$$

$$Y_p(0, y - mx)$$

$$PY_p = 1$$



$$\sqrt{x^2 + m^2x^2} = 1x^2 + m^2x^2 = 1m^2 = \frac{1-x^2}{x^2}$$

$$m = \pm \sqrt{\frac{1-x^2}{x^2}}$$

$$\frac{dy}{dx} = \sqrt{\frac{1-x^2}{x}}$$

$$\int dy = \int \frac{\sqrt{1-x^2}}{x} dx$$

Put $x = \sin \theta$

$$y = \int \frac{\cos^2 \theta}{\sin \theta} d\theta$$

$$\frac{dy}{dx} = -\frac{\sqrt{1-x^2}}{x}$$

$$\int dy = -\int \frac{\sqrt{1-x^2}}{x} dx$$

Put $x = \sin \theta$

$$y = \ln \left(\frac{1+\sqrt{1-x^2}}{x} \right) + c$$

$$\Rightarrow c = 0$$

$$\int \frac{1-\sin^2 \theta}{\sin \theta} d\theta = \int (\csc \theta - \sin \theta) d\theta = -\ln(\csc \theta + \cot \theta) + \cos \theta + c = -\ln \left(\frac{1+\sqrt{1-x^2}}{x} \right) + \sqrt{1-x^2}$$

Q15. Let b be a nonzero real number. Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function such that $f(0) = 1$.

If the derivative f' of f satisfies the equation $f'(x) = \frac{f(x)}{b^2 + x^2}$ for all $x \in \mathbb{R}$, then which of the following statements is/are TRUE ?

(A) If $b > 0$, then f is an increasing function

(B) If $b < 0$, then f is an decreasing function (C) $f(x) f(-x) = 1$ for all $x \in \mathbb{R}$

(D) $f(x) - f(-x) = 0$ for all $x \in \mathbb{R}$

Ans. A, C

Sol. f is non zero

$$\int \frac{f'(x)}{f(x)} dx = \int \frac{1}{x^2 + b^2} dx$$

$$\ln(f(x)) = \frac{1}{b} \tan^{-1} \frac{x}{b} + c$$

$$0 = 0 + c \{ \because f(0) = 1 \}$$

$$f(x) = e^{\frac{1}{b} \tan^{-1} \frac{x}{b}}$$

$$f(-x) = e^{\frac{1}{b} \tan^{-1} \frac{x}{b}}$$

(A) $f(x)$ is positive function for all $x \in \mathbb{R}$

$$f'(x) = \frac{f(x)}{x^2 + b^2}$$

(x) is also positive function

so $f(x)$ is increasing function

$$(d) f(x) - f(-x) = e^{\frac{1}{b} \tan^{-1} \frac{x}{b}} - e^{\frac{-1}{b} \tan^{-1} \frac{x}{b}} \neq 0$$

for all $x \in \mathbb{R}$

$$(c) f(x) f(-x) = 1$$

$$e^0 = 1$$

$$1 = 1$$

Q16. For any real numbers α and β let $y_{\alpha, \beta}(x)$, $x \in \mathbb{R}$, be the solution of the differential equation $\frac{dy}{dx} + \alpha y = x e^{\beta x}$, $y(1) = 1$. Let $S = \{ y_{\alpha, \beta}(x) : \alpha, \beta \in \mathbb{R} \}$. Then which of the following functions belong(s) to the set S ?

$$(A) f(x) = \frac{x^2}{2} e^{-x} + \left(e - \frac{1}{2} \right) e^{-x} \quad (B) f(x) = \frac{x^2}{2} e^{-x} + \left(e + \frac{1}{2} \right) e^{-x}$$

$$(C) f(x) = \frac{e^x}{2} \left(x - \frac{1}{2} \right) + \left(e - \frac{e^2}{4} \right) e^{-x} \quad (D) f(x) = \frac{e^x}{2} \left(\frac{1}{2} - x \right) + \left(e + \frac{e^2}{4} \right) e^{-x}$$

Ans. C, A

Sol. $\frac{dy}{dx} + \alpha y = x e^{\beta x}$

$$I. F. = e^{\int \alpha dx} = e^{\alpha x}$$

$$y \cdot e^{\alpha x} = \int e^{\alpha x} \cdot (x e^{\beta x}) dx + C$$

$$y \cdot e^{\alpha x} = \int_I x \cdot e^{(\alpha+\beta)x} dx + C$$

$$y \cdot e^{\alpha x} = \frac{x \cdot e^{(\alpha+\beta)x}}{(\alpha+\beta)} - \int \frac{e^{(\alpha+\beta)x}}{\alpha+\beta} dx + C$$

$$y \cdot e^{\alpha x} = e^{(\alpha+\beta)x} \left[\frac{x}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2} \right] + C$$

$$y(1) = 1$$

$$\Rightarrow e^\alpha = e^{\alpha+\beta} \left[\frac{1}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2} \right] + C$$

$$C = \left(1 - e^\beta \left(\frac{1}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2} \right) \right) e^\alpha$$

$$y \cdot e^{\alpha x} = e^{(\alpha+\beta)x} \left[\frac{x}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2} \right] + e^\alpha \left(1 - e^\beta \left(\frac{1}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2} \right) \right)$$

$$y = e^{\beta x} \left(\frac{x}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2} \right) + \left(e^\alpha - e^{\alpha+\beta} \left(\frac{1}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2} \right) \right) e^{-2x}$$

for $\alpha = \beta = 1$

$$y = \frac{e^x}{2} \left(x - \frac{1}{2} \right) + \left(e - \frac{e^2}{4} \right) e^{-x}$$

for $\alpha = 1, \beta = -1$

$$\frac{dy}{dx} + y = x \cdot e^{-x}$$

$$\text{If } e^{\int dx} = e^x$$

$$y \cdot e^x = \int x dx = \frac{x^2}{2} + C$$

$$y(1) = 1$$

$$e = \frac{1}{2} + C \Rightarrow C = e - \frac{1}{2}$$

$$y \cdot e^x = \frac{x^2}{2} + e - \frac{1}{2}$$

$$y = \frac{x^2}{2} \cdot e^{-x} + \left(e - \frac{1}{2} \right) e^{-x}$$

As (A, C)

- Q17. If $y(x)$ is the solution of the differential equation $xydy - (y^2 - 4y)dx = 0$ for $x > 0$, $y(1) = 2$, and the slope of the curve $y = y(x)$ is never zero, then the value of $10y(\sqrt{2})$ is _____.

Ans. 8

Sol. $xydy - (y^2 - 4y)dx = 0, x > 0$

$$\int \frac{dy}{y^2 - 4y} = \int \frac{dx}{x}$$

$$\int \left(\frac{1}{y-4} - \frac{1}{y} \right) dy = 4 \int \frac{dx}{x}$$

$$\log_e |y-4| - \log_e |y| = 4 \log_e x + \log_e c$$

$$\frac{|y-4|}{|y|} = cx^4 \xrightarrow{(1,2)} c = 1$$

$$|y-4| = |y|x^4$$

$$C-1$$

and

$$C-2$$

$$y-4 = yx^4$$

$$y-4 = -yx^4$$

$$y = \frac{4}{1-x^4}$$

$$y = \frac{4}{1+x^4}$$

$$y(1) = \text{ND (rejected)} \quad y(1) = 2$$

$$y(\sqrt{2}) = \frac{4}{5} \Rightarrow 10y(\sqrt{2}) = 8$$

- Q18. For $x \in \mathbb{R}$, let the function $y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} + 12y = \cos \left(\frac{\pi}{12} x \right), y(0) = 0. \text{ hen, which of the following statements is/are TRUE ?}$$

(A) $y(x)$ is an increasing function (B) $y(x)$ is decreasing function

(C) There exists a real number β such that the line $y = \beta$ intersects the curve $y = y(x)$ at infinitely many points

(D) $y(x)$ is a periodic function

Ans. C

Sol. $\frac{dy}{dx} + 12y = \cos\left(\frac{\pi x}{12}\right)$
 I.F. $e^{12x} \Rightarrow y \cdot e^{12x} = \int e^{12x} \cdot \cos\left(\frac{\pi x}{12}\right) dx + C$
 $\Rightarrow y \cdot e^{12x} = \frac{e^{12x}}{12^2 + \left(\frac{\pi}{12}\right)^2} \left[12 \cos \frac{\pi x}{12} + \frac{\pi}{12} \sin\left(\frac{\pi x}{12}\right) \right]$
 $\because y(0) = 0 \Rightarrow C = -\frac{12}{12^2 + \left(\frac{\pi}{12}\right)^2}$
 So $y = \frac{1}{\lambda} \underbrace{\left[12 \cos\left(\frac{\pi x}{12}\right) + \frac{\pi}{12} \sin\left(\frac{\pi x}{12}\right) \right]}_{f_1(x)} - 12e^{-12x}$
 $\frac{dy}{dx} = \frac{1}{\lambda} \underbrace{\left[-\pi \sin\left(\frac{\pi x}{12}\right) + \frac{\pi^2}{12^2} \cos\left(\frac{\pi x}{12}\right) \right]}_{f_2(x)} + 12e^{-12x}$

When x is large then $12e^{-12x}$ tends to zero.

But $f_2(x)$ varies in $\left[-\sqrt{\pi^2 + \left(\frac{\pi}{12}\right)^4}, \sqrt{\pi^2 + \left(\frac{\pi}{12}\right)^4} \right]$

Hence $\frac{dy}{dx}$ is changing its sign.

So $y(x)$ is non monotonic for all real number.

Also when x is very large then again $-12e^{-12x}$ is almost zero but $f_1(x)$ is periodic, so there exist some β for which $y = \beta$ intersect $y = y(x)$ at infinitely many points.

Q19. For $x \in \mathbb{R}$, let $y(x)$ be a solution of the differential equation

$(x^2 - 5) \frac{dy}{dx} - 2xy = -2x(x^2 - 5)^2$ such that $y(2) = 7$. Then the maximum value of the function $y(x)$ is

Ans. 16

Sol. $\frac{dy}{dx} - \frac{2x}{(x^2 - 5)} y = -2x(x^2 - 5)$
 $I f = e^{-\int \frac{2x}{x^2 - 5} dx} = e^{-\ln|x^2 - 5|} = \frac{1}{|x^2 - 5|}$
 Solution of DE $y \cdot \frac{1}{|x^2 - 5|} = \int \frac{-2x(x^2 - 5)}{|x^2 - 5|} dx + C$
 $\Rightarrow y \cdot \frac{1}{|x^2 - 5|} = -x^2 \left(\frac{x^2 - 5}{|x^2 - 5|} \right) + C$
 $y(2) = 7$
 $\Rightarrow 7 = 4 + C \Rightarrow C = 3$
 So $y = -x^2(x^2 - 5) + 3|x^2 - 5|$
 if $-\sqrt{5} < x < \sqrt{5}$ If $|x| > \sqrt{5}$

$$\begin{aligned}
 y &= (5 - x^2)(x^2 + 3) \\
 &= -x^4 + 2x^2 + 15 \\
 &= -[x^4 - 2x^2 - 15] \\
 &= -[(x^2 - 1)^2 - 16] \\
 &= 16 - (x^2 - 1)^2
 \end{aligned}$$

So max at $x = \pm 1$

$$\text{and } y_{\max} = 16$$

$$\begin{aligned}
 y &= (x^2 - 5)(-x^2 + 3) \\
 &= -x^4 + 8x^2 - 15 \\
 &= -[x^4 - 8x^2 + 15] \\
 &= -[(x^2 - 4)^2 - 1] \\
 &= 1 - (x^2 - 4)^2
 \end{aligned}$$

$$\text{Ans.} = 16$$

Exercise C1

- Q1. A function $y = f(x)$ satisfies the differential equation $f(x) \cdot \sin 2x - \cos x + (1 + \sin^2 x) f'(x) = 0$ with initial condition $y(0) = 0$. The value of $f(\pi/6)$ is equal to
(A) $1/5$ (B) $3/5$ (C) $4/5$ (D) $2/5$

Ans. D

Sol. -

- Q2. Water is drained from a vertical cylindrical tank by opening a valve at the base of the tank. It is known that the rate at which the water level drops is proportional to the square root of water depth y , where the constant of proportionality $k > 0$ depends on the acceleration due to gravity and the geometry of the hole. If t is measured in minutes and $k = \frac{1}{15}$ then the time to drain the tank if the water is 4 meter deep to start with is
(A) 30 min (B) 45 min (C) 60 min (D) 80 min

Ans. C

Sol. -

- Q3. The equation of the trajectories which is orthogonal to the family of curves $\cos y = ae^{-x}$ is
(A) $\sin y = c e^x$ (B) $\cos y = c e^x$ (C) $\sin y = c e^{-x}$ (D) None of these

Ans. C

Sol. -

- Q4. A function $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} - y = \cos x - \sin x$, with initial condition that y is bounded when $x \rightarrow \infty$. The area enclosed by $y = f(x)$, $y = \cos x$ and the y -axis in the 1st quadrant
(A) $\sqrt{2} - 1$ (B) $\sqrt{2}$ (C) 1 (D) $\frac{1}{\sqrt{2}}$

Ans. A

Sol. -

- Q5. A function $f(x)$ satisfying $\int_0^1 f(tx) dt = n f(x)$, where $x > 0$, is
(A) $f(x) = c \cdot x^{\frac{1-n}{n}}$ (B) $f(x) = c \cdot x^{\frac{n}{n-1}}$ (C) $f(x) = c \cdot x^{\frac{1}{n}}$ (D) $f(x) = c \cdot x^{1-n}$

Ans. A

Sol. -

Q6. A curve $y = f(x)$ passing through the point $\left(1, \frac{1}{\sqrt{e}}\right)$ satisfies the differential equation $\frac{dy}{dx} + xe^{-\frac{x^2}{2}} = 0$. Then which of the following does not hold good?

(A) $f(x)$ is differentiable at $x = 0$. (B) $f(x)$ is symmetric w.r.t. the origin.

(C) $f(x)$ is increasing for $x < 0$ and decreasing for $x > 0$.

(D) $f(x)$ has two inflection points.

Ans. B

Sol. -

Q7. A curve passing through $(2, 3)$ and satisfying the differential equation

$$\int_0^x ty(t)dt = x^2y(x), (x > 0) \text{ is}$$

(A) $x^2 + y^2 = 13$ (B) $y^2 = \frac{9}{2}x$ (C) $\frac{x^2}{8} + \frac{y^2}{18} = 1$ (D) $xy = 6$

Ans. D

Sol. -

Q8. A function $y = f(x)$ satisfies $(x+1) \cdot f'(x) - 2(x^2+x)f(x) = \frac{e^{x^2}}{x+1}$, $\forall x > -1$. If $f(0) = 5$, then $f(x)$ is

(A) $\left(\frac{3x+5}{x+1}\right) \cdot e^{x^2}$ (B) $\left(\frac{6x+5}{x+1}\right) \cdot e^{x^2}$ (C) $\left(\frac{6x+5}{(x+1)^2}\right) \cdot e^{x^2}$ (D) $\left(\frac{5-6x}{x+1}\right) \cdot e^{x^2}$

Ans. B

Sol. -

Q9. The solution of primitive integral equation $(x^2 + y^2)dy = xy dx$, is $y = y(x)$. If $y(1) = 1$ and $y(x_0) = e$, then x_0 is

(A) $\sqrt{2(e^2 - 1)}$ (B) $\sqrt{2(e^2 + 1)}$ (C) $\sqrt{3}e$ (D) $\sqrt{\frac{e^2 + 1}{2}}$

Ans. C

Sol. -

Q10. For the primitive integral equation $ydx + y^2dy = xdy$; $x \in \mathbb{R}$, $y > 0$, $y = y(x)$, $y(1) = 1$, then $y(-3)$ is

(A) 3 (B) 2 (C) 1 (D) 5

Ans. A

Sol. -

Q11. The solution of the differential equation $2x^2y \frac{dy}{dx} = \tan(x^2y^2) - 2xy^2$ given $y(1) = \sqrt{\frac{\pi}{2}}$ is

- (A) $\sin(x^2y^2) - 1 = 0$ (B) $\cos\left(\frac{\pi}{2} + x^2y^2\right) = 0$ (C) $\sin(x^2y^2) = e^{x-1}$
(D) $\sin(x^2y^2) = e^{2(x-1)}$

Ans. C

Sol. -

Q12. The general solution of the differential equation $\frac{dy}{dx} + x(x+y) = x(x+y)^3 - 1$ is

- (A) $\ln\left|\frac{(x+y+1)(x+y-1)}{(x+y)^4}\right| = x^2 + C$ (B) $\ln\left|\frac{(x+y+1)(x+y-1)}{(x+y)^2}\right| = x^2 + C$
(C) $2\ln\left|\frac{(x+y+1)(x+y-1)}{(x+y)^2}\right| = x^2 + C$
(D) $\ln\left|\frac{(x+y+1)(x+y-1)}{(x+y)^2}\right| = x + C$ (where C is arbitrary constant.)

Ans. B

Sol. -

Q13. The solution of the equation $\frac{dy}{dx} + \frac{1}{x}\tan y = \frac{1}{x^2}$ is

- (A) $2y = \sin y (1 - 2cx^2)$ (B) $2x = \cot y (1 + 2cx^2)$ (C) $2x = \sin y (1 - 2cx^2)$
(D) $2x \sin y = 1 - 2cx^2$

Ans. C

Sol. -

Q14. The equation of the curve such that the distance between the origin & the tangent line at an arbitrary point is equal to the distance between the origin & the normal at the same point is :

- (A) $x^2 + y^2 = c \cdot e^{\arctan(y/x)}$ (B) $\sqrt{x^2 + y^2} = c \cdot e^{-\arctan(y/x)}$ (C) $\sqrt{x^2 + y^2} = c \cdot e^{\arctan(y/x)}$
(D) $x^2 + y^2 = c \cdot e^{-\arctan(y/x)}$ where c is an arbitrary constant

Ans. B, C

Sol. -

Q15. Let $\frac{dy}{dx} + y = f(x)$ where y is a continuous function of x with $y(0) = 1$ and $f(x) = \begin{cases} e^{-x} & \text{if } 0 \leq x \leq 2 \\ e^{-2} & \text{if } x > 2 \end{cases}$. Which of the following hold(s) good?

(A) $y(1) = 2e^{-1}$ (B) $y'(1) = -e^{-1}$ (C) $y(3) = -2e^{-3}$ (D) $y'(3) = -2e^{-3}$

Ans. D, B, A

Sol. -

Q16. If $y(x)$ satisfies the differential equation $\frac{dy}{dx} = \sin 2x + 3y \cot x$ and $y\left(\frac{\pi}{2}\right) = 2$ then which of the following statement(s) is/are correct ?

(A) $y\left(\frac{\pi}{6}\right) = 0$ (B) $y'\left(\frac{\pi}{3}\right) = \frac{9 - 3\sqrt{2}}{2}$ (C) $y(x)$ increases in the interval

(D) $\int_{-\pi/2}^{\pi/2} y(x) dx = x$

Ans. C, A

Sol. -

Q17. Solve : $\tan y \frac{dy}{dx} + \tan x = \cos y \cdot \cos^3 x$

(A) $\sec x \sec y = \frac{1}{2} \left(x + \frac{1}{2} \sin 2x \right) + C$ (B) $((x + \sin x \cos x) + 2C) \cos x \cos y = 2$

(C) $\cos x \cos y = \frac{1}{2} (x + \sin x \cos x) + C$ (D) $\sec x \sec y = \frac{1}{2} (x + \cos x) + C$

Ans. A, B

Sol. -

Q18. The solution(s) of the differential equation $9(2 - y)^2 \left(\frac{dy}{dx} \right)^2 = 4(3 - y)$ is /are

(A) $(3 - x)^2 = y^2 (y + c)$ (B) $(x + c)^2 = y^2 (3 - y)$ (C) $y = 0$ (D) $y = 3$

Ans. D, B

Sol. -

Q19. Let $y = f(x)$ be a real valued function satisfying $x \frac{dy}{dx} = x^2 + y - 2$, $f(1) = 1$, then

(A) $f(x)$ is minimum at $x = 1$ (B) $f(x)$ is maximum at $x = 1$ (C) $f(3) = 5$ (D) $f(2) = 3$

Ans. A, C

Sol. -

Q20. Solution of differential equation $x \cos x \left(\frac{dy}{dx} \right) + y(x \sin x + \cos x) = 1$ is (where c is arbitrary constant)

(A) $xy = \sin x + c \cos x$ (B) $xy \sec x = \tan x + c$ (C) $xy + \sin x + c \cos x = 0$

(D) None of these

Ans. A, B

Sol. -

Q21. If $y = y(x)$ is solution of differential equation $(1 - x^2) \frac{dy}{dx} + xy = x$ such that $y(1) = 1$ and $g(x) = (y - 1)^2$, then

(A) $g(x)$ is an even function (B) $\lim_{x \rightarrow 1} \frac{g(x)}{(x - 1)^2} = 2$ (C) $g(x)$ has only two local minima

(D) $g(x) = 1$ has four solution

Ans. A, B, C, D

Sol. -